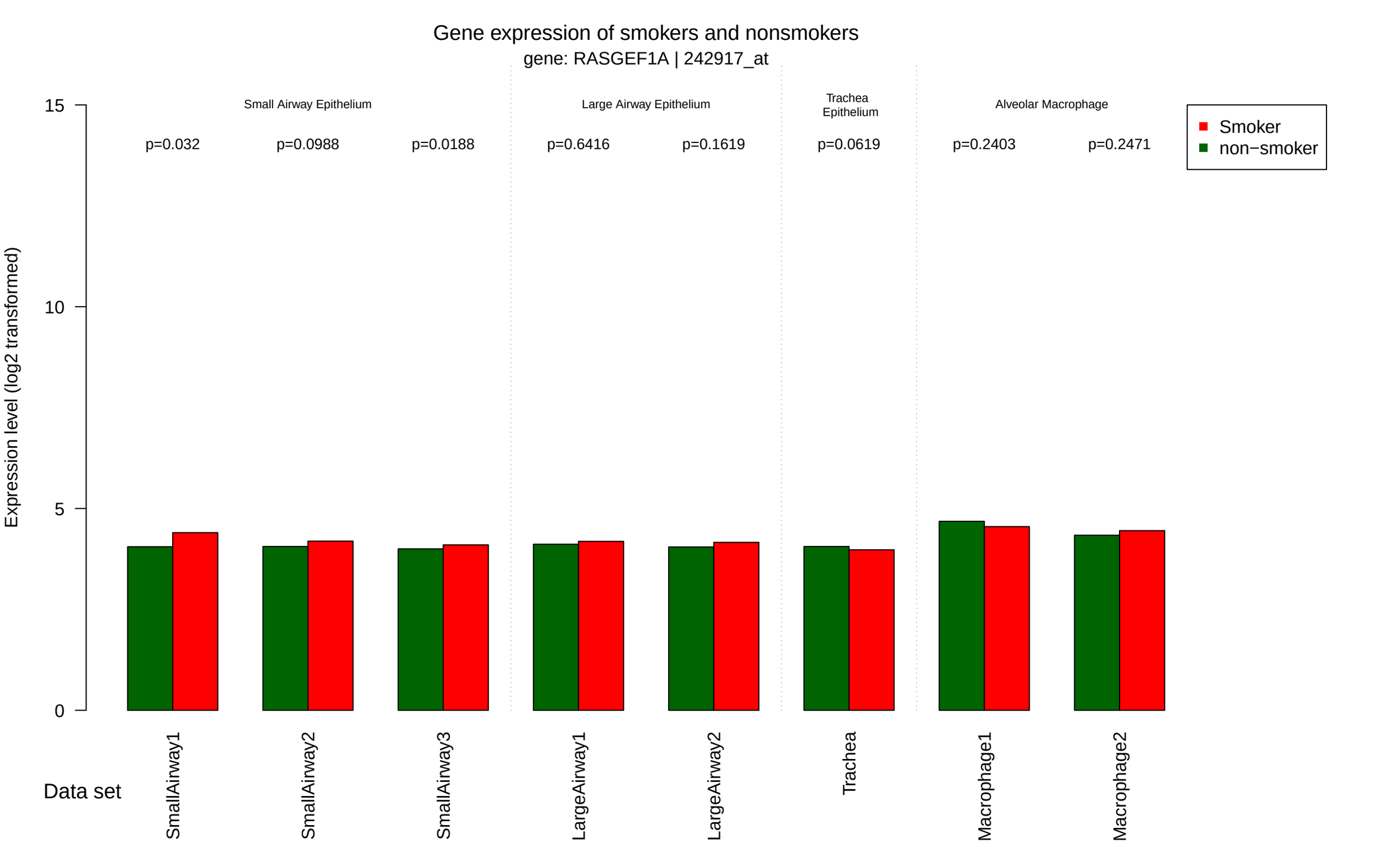
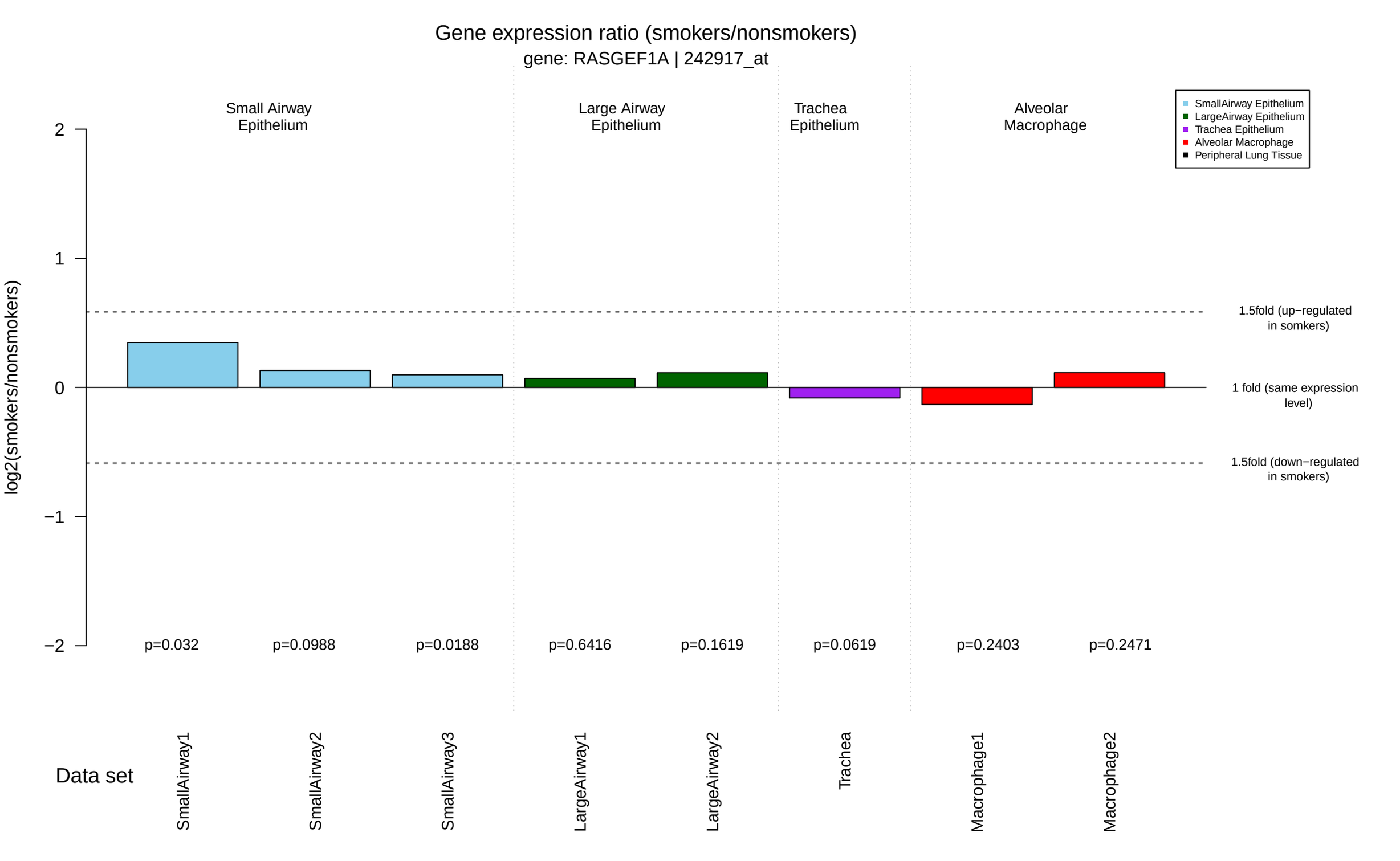
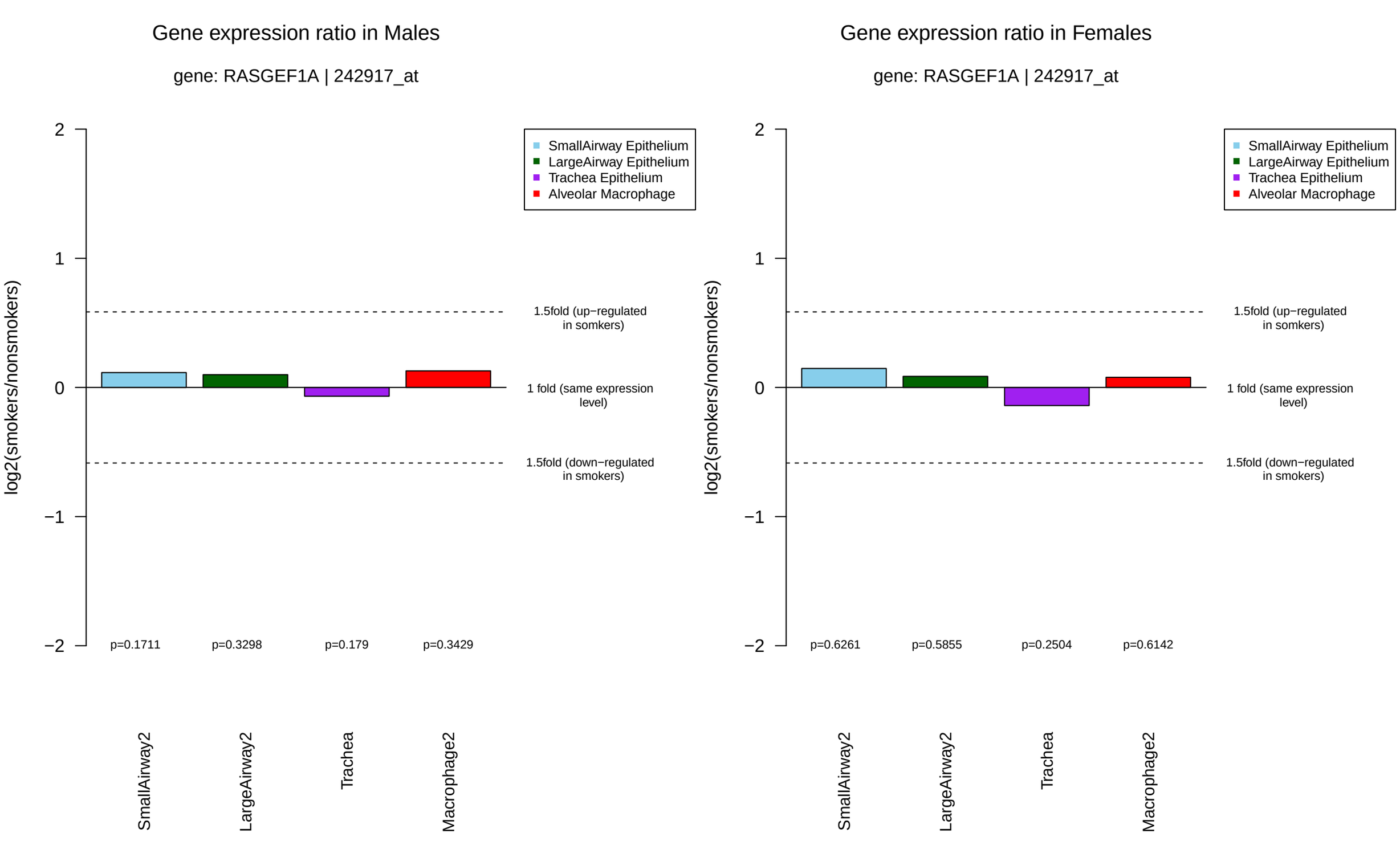
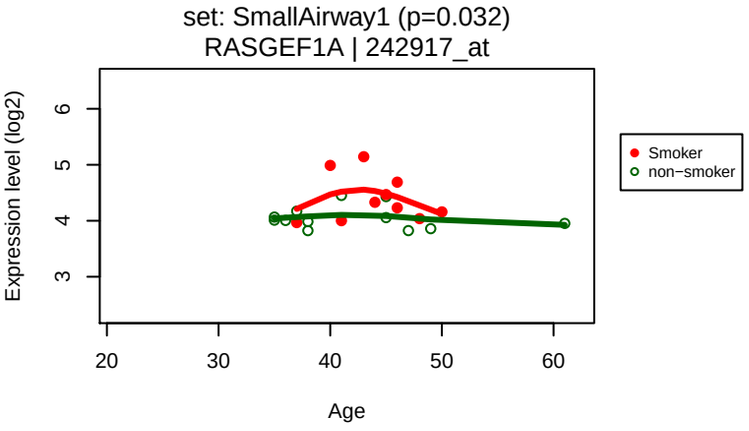
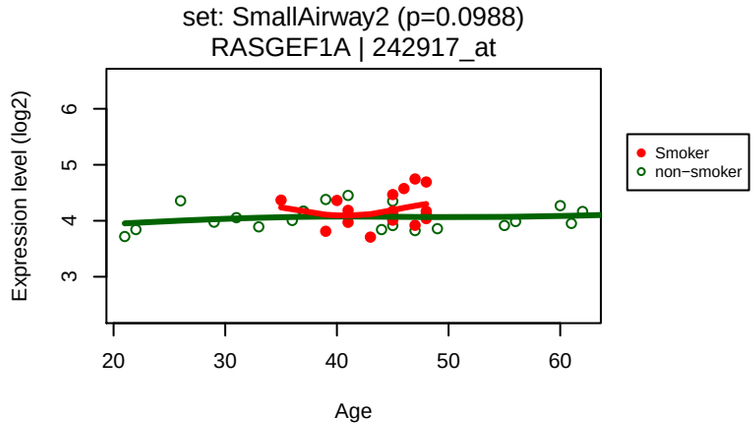
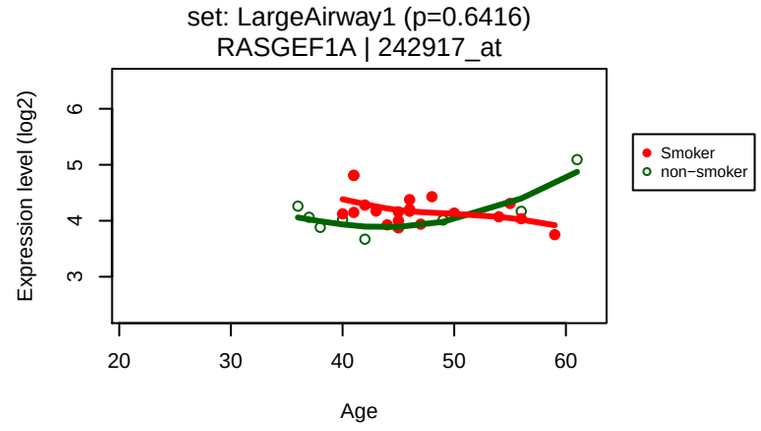

Gene expression of smokers and nonsmokers
gene: RASGEF1A | 242917_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.032
p=0.0988
p=0.0188
p=0.6416
p=0.1619
p=0.0619
p=0.2403
p=0.2471
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

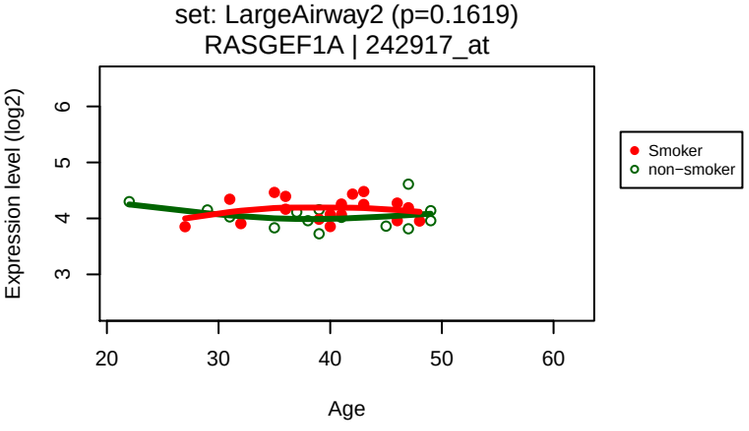

Gene expression ratio (smokers/nonsmokers)
gene: RASGEF1A | 242917_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.032
p=0.0988
p=0.0188
p=0.6416
p=0.1619
p=0.0619
p=0.2403
p=0.2471
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

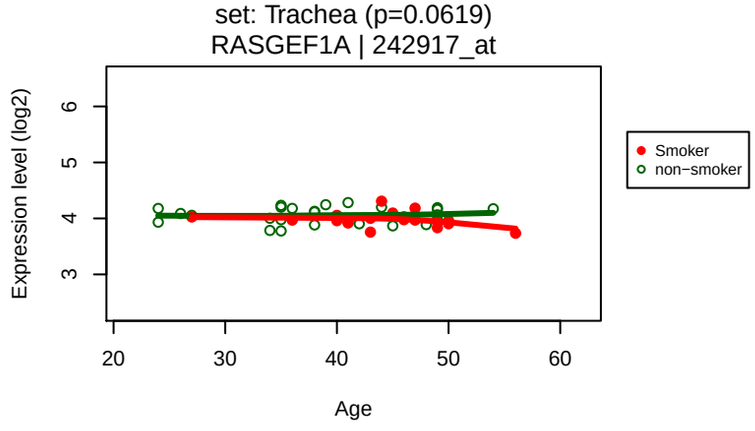

Gene expression ratio in Males
gene: RASGEF1A | 242917_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1711
p=0.3298
p=0.179
p=0.3429
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASGEF1A | 242917_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6261
p=0.5855
p=0.2504
p=0.6142
SmallAirway2
LargeAirway2
Trachea
Macrophage2

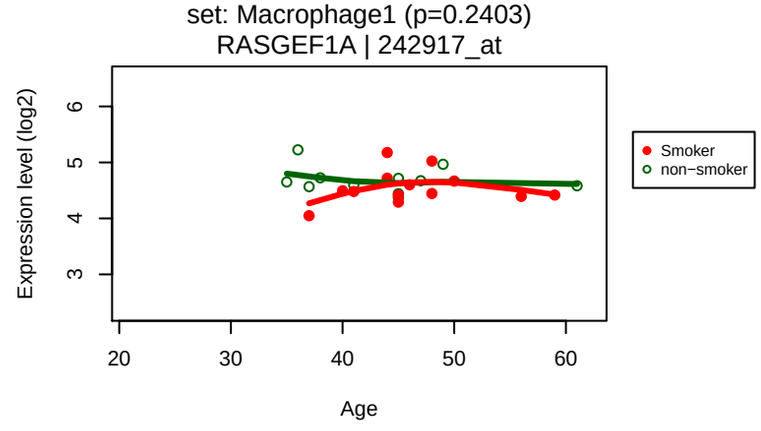

set: SmallAirway1 (p=0.032)
RASGEF1A | 242917_at
Expression level (log2)
Age
Smoker
non−smoker

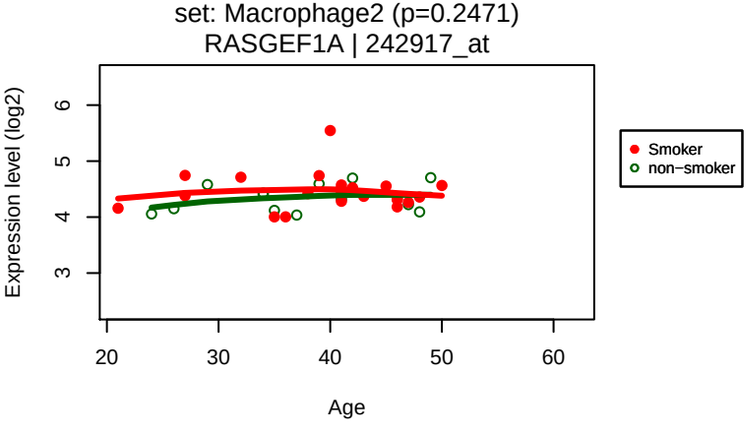

set: SmallAirway2 (p=0.0988)
RASGEF1A | 242917_at
Expression level (log2)
Age
Smoker
non−smoker

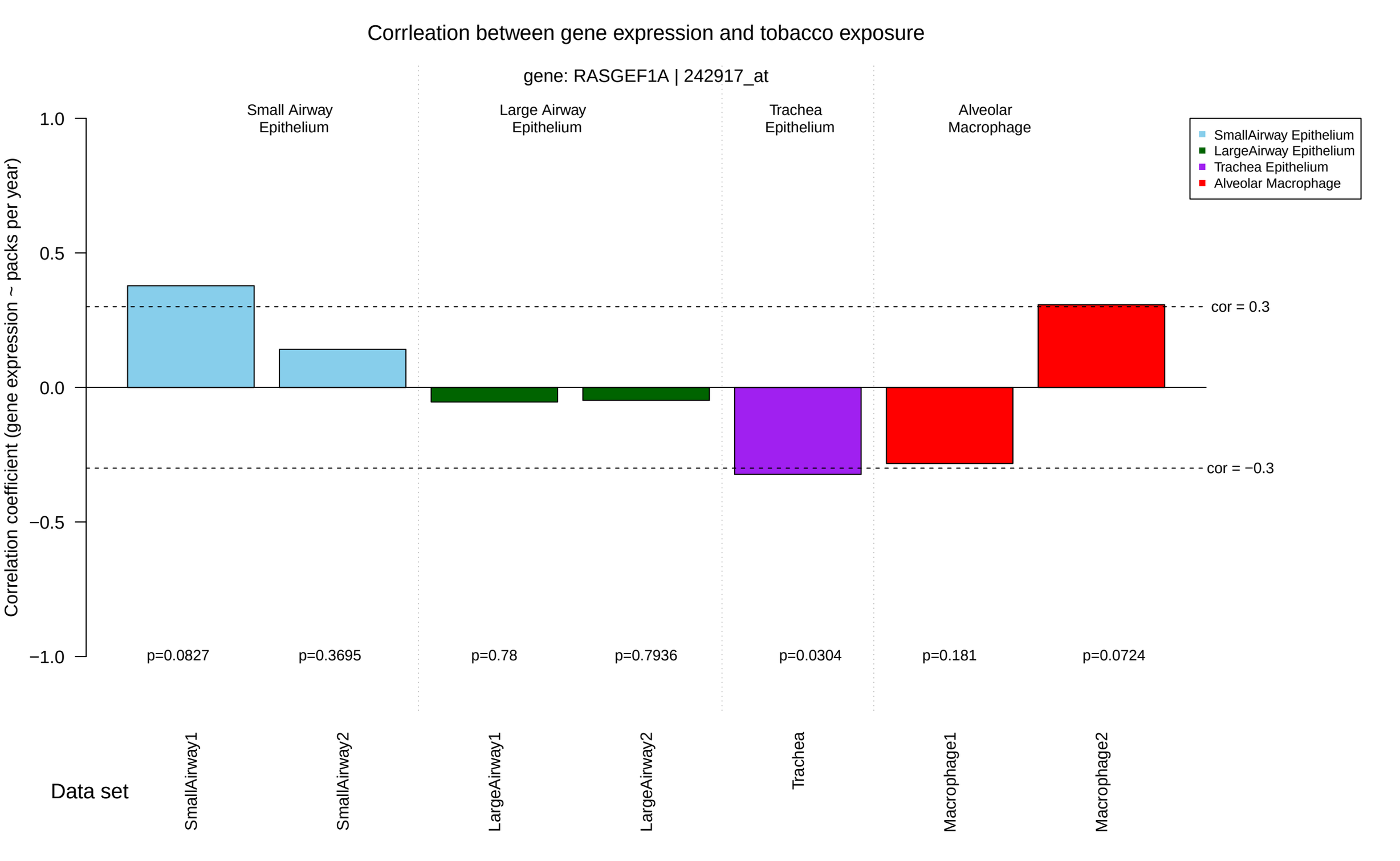

set: LargeAirway1 (p=0.6416)
RASGEF1A | 242917_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1619)
RASGEF1A | 242917_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0619)
RASGEF1A | 242917_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2403)
RASGEF1A | 242917_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2471)
RASGEF1A | 242917_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASGEF1A | 242917_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0827
p=0.3695
p=0.78
p=0.7936
p=0.0304
p=0.181
p=0.0724
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2